Cambridge (CIE) A Level Computer Science

Your notes

Assembly Language

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- Assembly Language Basics
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- Instruction Groups
- Addressing Modes

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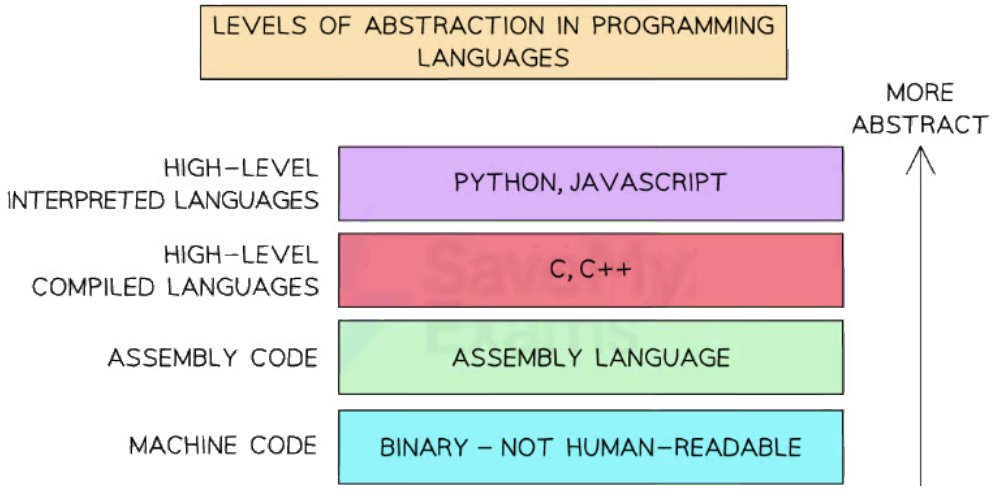
Machine code vs assembly

What is machine code?

- **Machine code** is a first-generation language
- Instructions are **directly executable by the processor**
- Written in **binary code**



Levels of Abstraction of Programming Languages

What is assembly?

- **Assembly code** is a second-generation language
- The code is written using **mnemonics**, abbreviated text commands such as **LDA (Load), STA(Store)**
- Using this language programmers can write **human-readable programs** that correspond **almost exactly to machine code**
- **One assembly language instruction** translates to **one machine code instruction**
- **Needs to be translated** into machine code for the computer to be able to execute it
- Each type of computer CPU has a **specific instruction set**

Instruction set

- An instruction set is a list of **all the commands** that can be **processed by a CPU**
- Each command has a **binary code** which is the **machine code**
- In assembly the binary code is written using mnemonics and split into an **opcode** and **operand**

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- The **opcode** is the **part of an instruction** that tells the CPU what **operation** to perform
- It's short for **"operation code"**
- Example operations: *LDA* (load), *STA* (store), *ADD*, *INP*, *OUT*, *BRZ*, *BRA*
- The **operand** is the **data** or **memory address** the opcode will work with
- It gives the **extra detail** the CPU needs to complete the instruction
- Often used to specify a **memory address**, a **value**, or a **label**

Opcode	Operand	Explanation
LDM	#n	Immediate addressing. Load the number n to the accumulator (ACC).
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC.
LDI	<address>	Indirect addressing. Use the value at the given address as a new address. Load contents to ACC.
LDX	<address>	Indexed addressing. Use <address> + index register to get the final address. Load to ACC.
LDR	#n	Immediate addressing. Load the number n to the index register (IX).
MOV	<register>	Move the contents of ACC to the given register (e.g. IX).
STO	<address>	Store the contents of ACC at the given memory address.
ADD	<address>	Add the contents of the given address to ACC.
SUB	<address>	Subtract the contents of the given address from ACC.
INC	<register>	Increment the contents of the register (ACC or IX) by 1.
DEC	<register>	Decrement the contents of the register (ACC or IX) by 1.
JMP	<address>	Unconditional jump to the given address.
CMP	<address>	Compare the contents of ACC with the contents of the given address.
CMP	#n	Compare the contents of ACC with the number n.

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CMI	<address>	Indirect addressing. Compare ACC with the contents at the address stored in <address>.
JPE	<address>	Jump to <address> if the previous compare result was **true** (equal).
JPN	<address>	Jump to <address> if the previous compare result was **false** (not equal).
IN		Take input from the keyboard and store its ASCII value in ACC.
OUT		Output to the screen the character stored in ACC.
END		End the program and return control to the operating system

All questions will assume there is only one general purpose register available (Accumulator)

ACC denotes Accumulator

IX denotes Index Register

<address> can be an absolute or symbolic address

# denotes a denary number, e.g. #123

B denotes a binary number, e.g. B01001010

& denotes a hexadecimal number, e.g. &4A

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Two-pass assembler

What is a two-pass assembler?

- A two-pass assembler **translates assembly language into machine code** in two stages
- It produces an **object program** that can be **stored**, **loaded**, and **run later**
- To execute this program, another utility called a **loader** is needed
- The assembler scans the source code **twice**:
  - In the **first pass**, it identifies all **labels** and records their memory addresses
  - In the **second pass**, it replaces those labels with the correct **memory addresses** in the machine code
- This **ensures the final machine code is accurate** and ready to be executed

Example

- The following program loads the value of A (5), adds the value of B (3), stores the result (8) in C, then halts:

```
    LOAD A
    ADD B
    STORE C
    HALT
A:  DATA 5
B:  DATA 3
C:  DATA 0
```

- **Pass 1:**
  - The assembler reads each line and assigns memory addresses to labels (e.g. A, B, C), but **does not translate the instructions yet**

Address	Instruction	Label	Add to symbol table
00	LOAD A		
01	ADD B		
02	STORE C		
03	HALT		
04	DATA 5	A	**A** = 04

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05	DATA 3	B	**B** = 05
06	DATA 0	C	**C** = 06

Symbol table created:

Label	Address
A	04
B	05
C	06

- **Pass 2:**
  - The assembler replaces labels with their corresponding memory addresses and outputs machine code
  - In this example we assume the following opcodes for simplicity:

Mnemonic	Opcode
LOAD	01
ADD	02
STORE	03
HALT	00
DATA	-

- Now generate the object (machine) code:

Address	Assembly	Machine code	Explanation
00	LOAD A	01 04	LOAD from address 04
01	ADD B	02 05	ADD value at address 05
02	STORE C	03 06	STORE into address 06

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03	HALT	00 00	Stop program
04	DATA 5	00 05	Data value 5
05	DATA 3	00 03	Data value 3
06	DATA 0	00 00	Data value 0 (placeholder)

Final object program:

```
01 04
02 05
03 06
00 00
00 05
00 03
00 00
```

Program tracing

What is a trace table?

- A **trace table** is a table used to **manually track the values of variables** as a program runs
- It helps to follow the flow of a program **line by line**, checking whether the logic works as expected
- Trace tables are used to:
  - **Understand how a program works**
  - **Debug errors** in the logic
  - **Predict outputs** before running the program
  - **Check variable values** at each step

Example

- An assembly program that loads the value from A (4) into the accumulator, adds the value from B (2), stores the result (6) into C, and halts

```
    LOAD A
    ADD B
    STORE C
    HALT
A:  DATA 4
B:  DATA 2
C:  DATA 0
```

- To trace the program, assumptions must be made:
  - **Accumulator (ACC)**: Used for calculations
  - Memory is addressed from 00 onwards

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- Simple instructions:
  - LOAD X → ACC = memory[X]
  - ADD X → ACC = ACC + memory[X]
  - STORE X → memory[X] = ACC
  - HALT → Stop

Trace table

Step	Instruction	ACC (Accumulator)	Memory[A]	Memory[B]	Memory[C]
0	(Start)	0	4	2	0
1	LOAD A	4	4	2	0
2	ADD B	6	4	2	0
3	STORE C	6	4	2	6
4	HALT	6	4	2	6

Final outcome:

- **ACC = 6**
- **C = 6**
- The program has successfully added A and B, then stored the result in C

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Categories of instructions

What are the categories of instructions?

- Assembly language instructions can be grouped into:
  - **Data movement**
  - **Input and output of data**
  - **Arithmetic operations**
  - **Unconditional and conditional instructions**
  - **Compare instructions**
- For each category, a table containing the instruction (Opcode and Operand), and an explanation is given

Category	Opcode	Operand	Explanation
Data Movement	LOAD	A	Loads the value stored at memory location A into the accumulator.
	STORE	B	Stores the value from the accumulator into memory location B.
	MOV	A, B	Copies the value from register/memory A to register/memory B.
	CLEAR	ACC	Clears the accumulator (sets it to 0).
	DATA	5	Declares a constant or data value (e.g. DATA 5 stores the value 5).
Input and Output of Data	IN		Takes input from a user or input device and stores it in the accumulator.
	OUT		Outputs the value in the accumulator to a screen or output device.
Arithmetic Operations	ADD	A	Adds the value at memory location A to the accumulator.

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	SUB	B	Subtracts the value at memory location B from the accumulator.
	INC		Increments the accumulator by 1.
	DEC		Decrements the accumulator by 1.
Unconditional and Conditional	JMP	LABEL	Unconditionally jumps to the instruction at LABEL.
	JZ	LABEL	Jumps to LABEL if the accumulator is zero.
	JNZ	LABEL	Jumps to LABEL if the accumulator is **not** zero.
	JC	LABEL	Jumps to LABEL if the carry flag is set.
	HLT		Stops the program (halts execution).
Compare Instructions	CMP	A	Compares the accumulator with the value at memory location A.
	TEST	A	Performs a logical AND between accumulator and A; sets flags, no output.

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Addressing methods

What is an addressing method?

- An addressing method is a way in which **an instruction in assembly language or machine code** can **access data stored in memory**
- There are five main addressing methods:
  - **Immediate**
  - **Direct**
  - **Indirect**
  - **Indexed**
  - **Relative**

Addressing mode	Description	Example instruction	Explanation
Immediate	The **operand is a constant value** included directly in the instruction	MOV AX, 1234h	Moves the **immediate value** 1234h into the AX register
Direct	The **exact memory address** of the operand is given in the instruction	MOV AX, [1234h]	Moves the value **stored at memory location** 1234h into the AX register
Indirect	A **register contains the memory address** of the operand	MOV AX, [BX]	If BX = 2000h, this moves the value **at memory location** 2000h into AX
Indexed	Uses a **base register and an index** to calculate the memory address	MOV AX, [BX + SI]	If BX = 0050h and SI = 1000h, this moves the value **at** 1050h into the AX register
Relative	The operand is an **offset** relative to the **current instruction address**	JMP +5	Tells the program to **jump forward 5 instructions** from the current line Used in branching

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